PRICING SHOULDN'T BE A GUESS.

Dynamic Pricing & Revenue Management Platform

A Business Discovery & Market Research Report

Why travel & cruise companies are replacing spreadsheets with a rules-driven pricing engine — and what it will take to build one.

DOCUMENT 1 OF 4

Business Discovery → Requirements → System Design → Implementation Plan

START HERE

A Quick Note Before You Dive In

This report answers one question: **why does a cruise or travel company need a custom pricing platform, instead of just using a spreadsheet or an off-the-shelf tool?** It's written to be skimmed in fifteen minutes or studied in an hour — diagrams first, details second.

> ■ **CURIOUS?**
>
> *If a hotel room and a laptop are both "products," why can't they be priced the same way? Keep this question in mind — the answer explains almost everything in this report.*

WHAT'S INSIDE

01 · THE FOUNDATION

The Perishable Problem

Why travel inventory behaves nothing like a normal product

Retail products wait. A laptop that doesn't sell today can sell next month at the same price. Travel inventory doesn't get that privilege. A cruise cabin, a hotel room, or a flight seat has a hard expiry: the departure date. Once that date passes, an empty cabin isn't discounted — it's worthless.

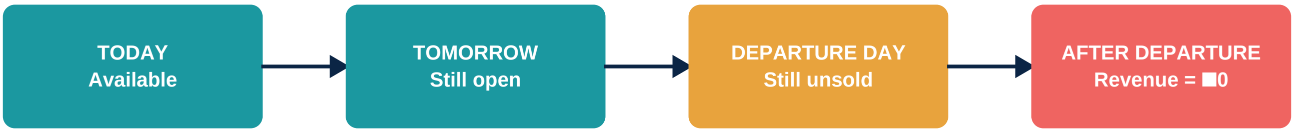

Figure 1 — A cruise cabin's value collapses to zero the moment the ship departs.

This single fact — **inventory expires** — is the entire reason revenue management exists as a discipline. Airlines discovered it first during deregulation, and the same logic has since spread to hotels, cruise lines, car rentals, and online travel agencies.

0	**100%**	**1**
Value of an unsold cabin after departure	Perishable — no inventory carries over	Chance to sell it before it's gone

A fixed price ignores this reality completely. A cabin priced at ■18,000 stays at ■18,000 whether it's a slow Tuesday or a festival weekend running at 95% occupancy — leaving real money on the table in one direction, and empty cabins in the other.

> ■ **CURIOUS?**
>
> *A cabin sitting at 95% occupancy three days before departure — should it get more expensive, or cheaper, to fill that last 5%? Static pricing can't even ask the question.*

What Is Revenue Management?

And how it differs from the "dynamic pricing" everyone talks about

Revenue Management is the discipline of selling the **right product**, to the **right customer**, at the **right time**, through the **right channel**, for the **right price** — with the goal of maximizing revenue while minimizing unsold inventory. Dynamic pricing is simply one of its execution tools.

	Revenue Management	Dynamic Pricing
Nature	Business strategy	Pricing technique
Focus	Overall revenue & inventory	Changing individual prices
Horizon	Long-term optimization	Short-term, real-time
Includes	Forecasting, inventory, pricing	Real-time price adjustment

A related idea, **yield management**, is Revenue Management applied to a fixed, perishable inventory pool — e.g. "sell all 100 cabins on this sailing for the highest possible total revenue."

MULTI-CHANNEL COMPLEXITY

Companies rarely sell through just one door. Every channel below may need its own price, because each carries different commissions and contract terms.

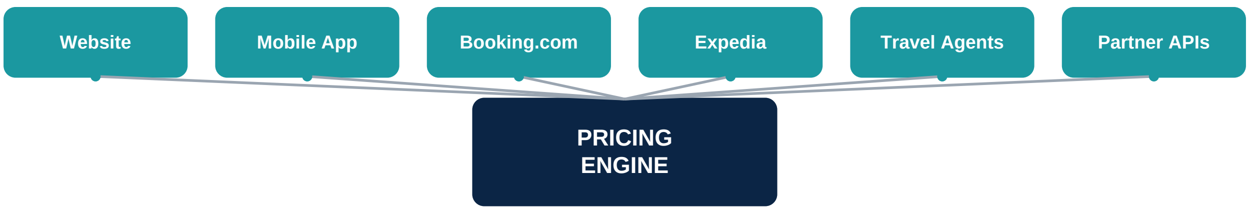

Figure 2 — One pricing engine has to speak to every sales channel at once.

WHAT ACTUALLY MOVES A PRICE

- Seasonality & holidays
- Booking window (days to departure)
- Customer segment
- Promotions & events
- Occupancy remaining
- Competitor prices
- Sales channel
- Market demand signals

03 · THE DIAGNOSIS

Where Today's Process Breaks

Twelve symptoms, all tracing back to one root cause

Most travel companies still run pricing through spreadsheets, scattered emails, and individual judgment calls. That process breaks down in the same handful of ways, again and again:

Problem	Business Impact	What Fixes It
Manual pricing	Slow, error-prone updates	Rule-driven automation
Static pricing	Missed peak-demand revenue	Demand-based dynamic pricing
Revenue leakage	Selling too cheap, too late	Automated, timely rule execution
Unsold inventory	Cabins/seats expire worthless	Occupancy-aware price shifts
Multi-channel chaos	Conflicting prices across OTAs	One centralized pricing engine
Slow market response	Can't react to a sudden sale	Real-time rule publishing
No central rule store	Knowledge lives in people's heads	Single source of pricing truth
No simulation	Rules go live untested	Preview pricing before publish
No personalization	Same price for every customer	Segment- & loyalty-based pricing
Poor audit trail	“Why did this price change?”	Full change history & logs

> ■ CURIOUS?
>
> *Notice that every “fix” column points to the same idea — a centralized, rule-driven system. That's not a coincidence; it's the thesis of this whole report.*

04 · THE MARKET SCAN

What Already Exists — and Its Limits

Commercial Revenue Management Systems are mature. So are their constraints.

Platform	Primary Focus	Key Strength
IDeaS	Hotels, airlines	AI forecasting & optimization
Duetto	Hotels & resorts	Open, channel-level pricing
Lighthouse	Hotels	Market intelligence
Atomize	Hotels	Fully automatic AI pricing
Pricepoint	Hotels (SMB)	Accessible dynamic pricing

These tools do dynamic pricing, demand forecasting, and reporting well. But they're built for a broad customer base — which means they struggle with cruise-specific workflows, unique partner contracts, and internal approval processes that don't match the standard hotel playbook.

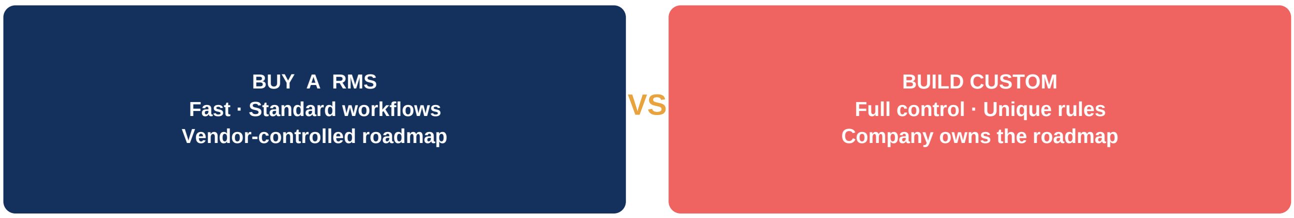

Figure 3 — Buying gets you speed; building gets you control. Most enterprises eventually need both.

A CUSTOM RULE MIGHT LOOK LIKE THIS

IF Platform = Travel Agent A **AND** Customer = Corporate **AND** Cabin = Luxury Suite **AND** Departure < 15 Days → **Increase Price by 18%**
ELSE → Apply Corporate Contract Rate

Off-the-shelf RMS platforms rarely support this exact combination of conditions without heavy, costly customization — which is precisely the gap a custom platform closes.

Vision, Value, and a Three-Phase Roadmap

A custom platform isn't built because commercial software is bad — it's built because every travel company's pricing rules, partner contracts, and approval flows are different. The product vision is a single, centralized system where business teams configure, simulate, and publish pricing — without writing code.

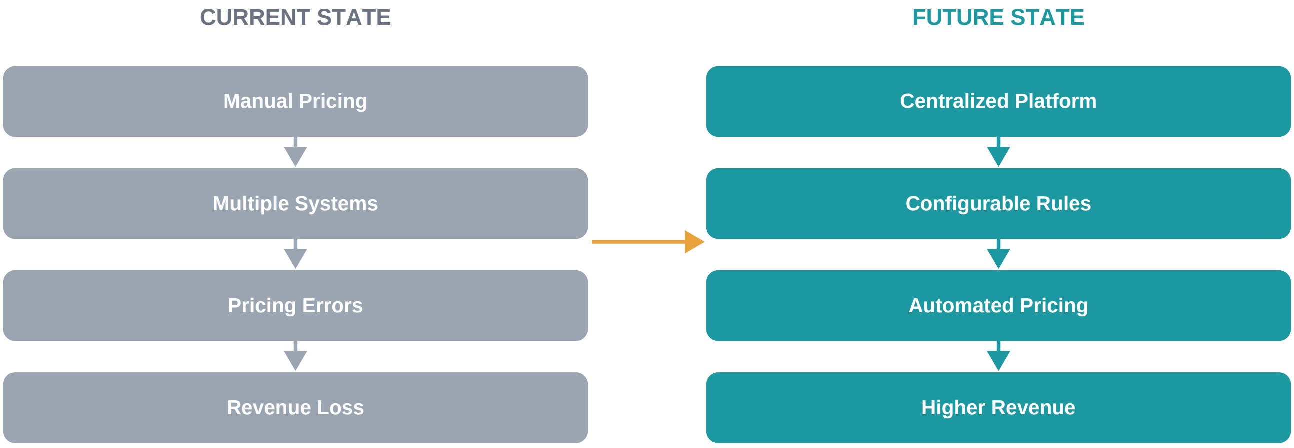

Figure 4 — The shift from fragmented manual pricing to one governed, automated system.

A ROADMAP, NOT A BIG-BANG LAUNCH

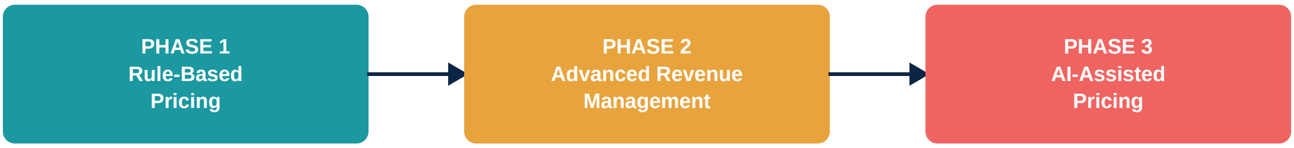

Figure 5 — The platform is designed to mature in three phases — rules first, forecasting second, AI third.

Phase 1 delivers configurable rule-based pricing across channels. Phase 2 adds demand and occupancy forecasting plus analytics. Phase 3 introduces AI-assisted recommendations, competitor analysis, and automated optimization.

What the Platform Must Do

Sixteen capabilities, in priority order

Capability	Priority	What It Delivers
Rule Management	High	Create, edit & schedule pricing rules — no developers needed
Seasonal Pricing	High	Different prices for summer, winter, peak & off-season
Holiday & Event Pricing	High	Configurable pricing for festivals & major events
Occupancy-Based Pricing	High	Price rises automatically as inventory fills
Booking-Window Pricing	High	Early-bird discounts → last-minute premiums
Customer Segment Pricing	High	Loyalty, VIP, corporate & agent rates
Sales Channel Pricing	High	Website, OTA & partner-specific pricing
Promotional Pricing	High	Scheduled flash sales & festival discounts
Rule Priority Management	High	Predictable outcomes when rules overlap
Price Simulation	High	Preview the final price before publishing
Audit Trail	High	Who changed what, when, and why
API Integration	High	Booking site, app & partner systems consume prices
Scalability	High	Many ships, routes, cabins & currencies
Rule Scheduling	Medium	Future-dated activation, no manual trigger
Manual Price Override	Medium	Authorized exceptions for VIP/corporate cases
Reporting & Analytics	Medium	Revenue, occupancy & promotion performance

Occupancy-based pricing, as one example, might look like this in practice:

Figure 6 — As occupancy climbs, price increases accelerate to protect the remaining inventory.

07 · THE TAKEAWAY

The Real Insight: This Isn't a Pricing Script

One detail from stakeholder conversations reframes the entire project. The requirement wasn't “build a pricing algorithm” — it was: *“cruise company needs custom logic, platform-specific pricing management module/dashboard.”*

THE CORE PRODUCT ISN'T THE ALGORITHM — IT'S THE RULE ENGINE

The real product is a **Business Rules Management Platform (BRMS) for Pricing**: a system that lets business teams define *how* prices should be calculated — without writing code. Dynamic pricing becomes just one category of rule the engine can run, not the whole architecture.

That distinction changes everything downstream. Instead of hardcoding pricing logic, the technical architecture (covered in Documents 2–4) is designed around a configurable rules engine — making the platform extensible to promotions, approvals, and future AI-driven logic without a rebuild.

■ CURIOUS?

If pricing is just one type of rule, what else could this engine eventually govern — approval workflows? Refund policies? Loyalty tiers? That's the extensibility this architecture is built for.

COMING NEXT

Document 2 (Functional Requirements), Document 3 (System Design), and Document 4 (Technical Implementation Plan) translate this business case into the rule-engine architecture, data flow, APIs, and phased build plan.